



Introduction to Artificial Intelligence (INTRAI)

Module 2: How AI Systems Work

Examples and Data Quality

Lesson Notes

KMVTECHUG

Version: 2026

Everyday Examples

Spam filter: data is past emails labelled spam or not; prediction is whether a new email looks like spam.

Face unlock: data is your enrolled face images; prediction is match or no match.

Route suggestion: data is traffic and map patterns; prediction is a faster path.

Mobile-money risk score: data is past transaction patterns; prediction is an unusual activity score.

Same pipeline, different goals and data.

Why Data Quality Matters

Garbage in, garbage out. If labels are wrong, rare groups are missing, or photos are only from one region, the model learns a skewed picture.

A crop disease model trained only on one district's photos may fail on different lighting, soil, or leaf varieties elsewhere in Uganda.

Asking about data source and coverage is a professional skill, not only a technical one.

Three Check Questions

What data trained this tool?

What does it output?

What happens when the situation differs from training?

Reflect: Which of the three check questions would you ask first about a new campus chatbot?

Closing

Module 2 core idea: AI systems turn data into models that make predictions for a defined goal.
Module 3 introduces machine learning types in plain language.